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The Influence of Intrinsic Motivation on Decisions Between Actions

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Abstract

Extensive research explains how pre-frontal cortical areas process explicit rewards, and how pre-motor and motor cortices are recipients of that processing to energize motor behaviour. However, the specifics of motor behaviour, decisions between actions and brain dynamics when driven by no explicit reward, remain poorly understood. Are patterns of decision and motor control altered wen performing under social pressure? Are the same brain regions that typically process explicit rewards also involved in this expression of motivation? To answer these questions, we designed a novel task of decision-making between precision reaches and manipulated motivation by means of social pressure, defined by the presence or absence of virtual partner of a higher/lower aiming skill than our participants. We assessed the overall influence of this manipulation by analysing movements, decisions, pupil dilation and electro-encephalography. We show that the presence of a partner consistently increased aiming accuracy along with pupil diameter, furthermore the more skilled the partner. Remarkably, increased accuracy is attained by faster movements, consistently with a vigour effect that breaches speed-accuracy trade-offs typical of motor adaptation. This implicated an ensemble of cortical sources including pre-frontal areas, concerned with the processing of reward, but also pre-motor and occipital sources, consistent with the nature of the task. Overall, these results strongly suggest the role of social pressure as a motivational drive, enabling an increase of both vigour and accuracy in a non-trivial fashion.

Introduction

Motivation has been often studied within paradigms in which rewarding stimuli elicit subsequent behaviour, assessing motivation through proxies of physiological variables[1–3]. For example, the effect of explicit reward on action invigoration[4,5], preparatory attention, working memory and decision-making[6–9], identified multiple pre-frontal cortical regions that encode specific aspects of reward processing[3,10–13]. However, anecdotal evidence suggests that, beyond explicit rewards, the brain's motivational systems evolved to deal with situations in which reward could be either non-explicit or just absent, e.g., animals tends to explore their environment during minutes or hours driven by motivation alone, in the prospect of finding food. Consistent with this, earlier theories defined motivation as *a substance, held back in a container, capable of energizing behaviour, and subsequently released in action*[14,15]. In other words, motivation is viewed as a midfield physiological actor, sensitive to reward and subliminal priming[16], which releases through behaviour. Fifty years after Bindra's definition, our understanding of the processes underlying motivation in the absence of external reward remain elusive. Despite motivation's covert nature, physiological proxies have shown that motivation decreases when reward is obtained, contingent on performance[17,18], and that it increases with behavioural activation[19], encompassed by connectivity modulations between the right medial frontal gyrus and the right posterior cingulate cortex. Likewise, a differential manipulation of extrinsic reward and intrinsic motivation in an imaginary task identified the angular gyrus[20] and the anterior insular cortex[21], as their respective regions of interest[22]. In summary, although we are progressively gathering a remarkable amount of information about motivational

processes, our current characterization of motivational dynamics per se, within the brain and its relationship to motor behaviour in the absence of external rewards, remains clearly incomplete.

One of the main remaining questions is whether the degree of differentiation between motivational subsystems as a function of whether reward is explicitly presented[23]. Likewise, is the contribution of external reward incremental on the overall neural dynamics of how motivation relates to behaviour? How are the neuronal circuits concerned with motivation alone localized or distributed across the brain? To answer these questions, we sought an indirect or subliminal source of motivation, such as social pressure, which may sometimes exert a profound effect on behaviour[24,25], and performed a behavioural-cortical characterization of the influence of intrinsic motivation onto motor behaviour and decision-making. To this end, we designed a novel decision-making task where human participants had to select between two precision reachings, and manipulated motivation by social pressure, in the absence of an explicit external reward. In brief, each participant played either alone or accompanied by a virtual partner of a lesser/higher aiming skill than him/her. We recorded arm kinematics, decisions, high density electro-encephalograms, and pupil diameter from a set of healthy participants. Our goal was dual; first, to provide a formal characterization of the neural dynamics and pupil diameter dynamics of healthy subjects underlying this modulation; second, to identify the correlates of decisions and selected movement parameters with the social hierarchy. Our analyses of movement showed that, for each level of social pressure, the choice of motor parameters and decisions were consistent with a concern to rank higher than their partner, encompassing any increase of the partner's skill and of pupil diameter. Furthermore, electrode and source level analyses of activity and connectivity of the participant's electro-encephalograms (EEGs), showed that specific pre-frontal and pre-motor areas are responsible for this modulation of behaviour, and therefore are best candidates for a cortical network of intrinsic motivation.

Results

Our goal was to examine how social pressure modulates internal motivation in an experiment of decision-making between precision reachings. To this end, we analyzed its subsequent effect on aiming accuracy, on the choice of movement parameters, and on the dynamics of the related cortical network. Eleven participants were instructed to freely select one of two planar, curved reaching movements of opposite biomechanical cost[26,27], and to hit the centre of their target of choice (METHODS; FIG 1B-C). We define the manipulation of our experimental Motivation per block of trials as a function of whether the participant would play alone (M=0, Solo), or accompanied by a virtual partner, either less (M=1; Easy) or more skilled (M=2; Hard) than our participant. During the accompanied blocks the partner would perform the same sequence of trials; and we showed their respective accuracies at the end of each trial, and average accuracy ranking each nine trials (see METHODS; FIG 1A). Two kinds of simulated partners were generated, with an average aiming accuracy lower/higher than our participant (see METHODS; FIG 1D). Importantly, we told the participant that the partner's purpose was to provide companionship, and that he/she was not competing. Any such mention to competition during the task was denied and discouraged. In addition to Motivation, blocks were also classified as a function of Control Regime: *stop-*

in (requiring the reaching movement to stop at the target), and *cross-over* (hitting and crossing over the target)[28] — FIG 1C. In both cases, our hypothesis is that social pressure motivates our participants to increase precision, and to consequently adjust their decisions and principles of selection of motor parameters (see conceptual framework described in FIG 1E). We recorded movement kinematics and pattern of decision from movement data, ocular and electro-encephalographic data.

Kinematic Correlates of Motivation

The purpose of the two control regimes is having two conditions in which the requirement of control as a function of motor noise[29,30,31]. In both cases, we predicted that our modulation of motivation could act in one of two ways: either participants were sensitive to social pressure per se and reduced their aiming error regardless of relative aiming skill, or they were concerned by the social ranking and reduced their accuracy only when the partner's accuracy was better than the participant's. In either case, if the adaptive strategy were consistent with a speed-accuracy trade-off, the error reduction should extend deceleration times during stop-in trials and peak velocities during cross-over ones. In other words, increased precision during stop-in trials should correlate with longer movements and weaker late movement kinematics, from Peak Velocity (PV) to the end of the movement at target arrival. By contrast, cross-over movements enter the target around PV, which make any strategy to control precision contingent on modulating early kinematic markers, from the onset of movement to PV. Detailed analyses for each control regime are shown next (FIG 2 & Suppl. FIG 1).

We first assessed the influence of motivation on *stop-in* movements (FIG 2D) by fitting the distributions of middle and late trajectory kinematic markers (z-scored Peak Velocity —PV, Peak Deceleration —PD, Time-To-Peak-Velocity — TTPV and Deceleration Time —DT, see METHODS) with a generalized linear model (GLM) per participant (see METHODS; FIG 2A-B). The GLM included three main regressors: Motivation (M in the FIG 2, with values 0-Solo, 1-Easy, 2-Hard), two additional factors that quantify habituation (#Block, #Trial within block); and the interaction of biomechanics and target choice (see METHODS; BM×C, 0-Low Cost/1-High Cost), which captured the influence of motor cost (FIG 2A). We also included the interactions of Motivation, #Block, #Trial and BM×C. Group significance of the regression parameters was established with the non-parametric Mann-Whitney-Wilcoxon test reporting the z-value for the effect size in addition to the p-value [32], by comparison with a ten-thousand sample surrogate distribution for each regression coefficient (FIG 2A-B). This surrogate distribution was obtained by fitting the GLM to shuffled kinematic marker values across trials per participant, which were then pooled across all participants.

Consistent with prediction, the arrival error diminished with increasing motivation ($p=0.00071$; $z=-3.38$), encompassed by a reduction of PV ($p=0.017$; $z=-2.37$) and PD ($p=0.019$; $z=2.34$), and of TTPV ($p=0.0012$; $z=-3.21$) and DT ($p=0.028$; $z=-2.19$) intervals (FIG 2B). Also, accuracy co-varied with velocity and movement duration when performing Solo. However, the joint PV & TTPV reduction for motivated states (M=1-Easy, 2-Hard) did not exhibit any significant correlation with error (FIG 2E), thus violating the speed-accuracy trade-off when performing with a partner. In other words, although both the error and PV

diminished within motivated states 1 and 2, they were uncorrelated. Along with Motivation, the error also diminished as the experimental session unfolded, with #Block (p=0.00067; z=-3.40; FIG 2F), also encompassed by a marginally non-significant reduction of PV (p=0.080; z=1.74) and PD (p=0.073; z=-1.79), possibly by the gradual increase of efficiency. The dependence on Motivation and adaptiveness of the participant's responses was also corroborated by the interaction Motivation × #Block (p=0.0052, z=2.79).

Supplemental FIG 1 shows the effect of social pressure on the motor control of *cross-over control regime* (Suppl. FIG 1H). Previous evidence suggested two relevant behavioral differences with respect to stop-in: first, target arrival when crossing-over is necessarily more difficult, as this occurs around peak velocity, the time of maximum vigor and motor noise of the movement[29]; second, the piece of the movement beyond target crossing is inconsequential to arrive to a precise target location. Our hypothesis was that a lower social categorization (being ranked second best in motivated states 1 or 2) positively motivates the participant to increase precision by decreasing the intensity of the earlier two metrics of movement: Peak Velocity (PV) and Time-To-Peak-Velocity (TTPV) — see Methods. Our results yielded that the arrival error was mildly reduced with increasing motivation (p=0.026; z=-2.22), encompassed by an elongation of the movement time to target — TTPV (p=0.036; z=2.10). Furthermore, much like in the stop-in regime, the reaching strategy did significantly improve with #Block by reducing the error (p=0.021; =-2.22), PA (p=0.027, z=-2.21) and PV (p=0.0083; z=-2.64) — Suppl. FIG 1D. However, unlike for the Stop-In case, the speed-accuracy trade-off relating end-point error and PV and TTPD held across all conditions (Suppl. FIG 1E). In other words, if the control requirements were low (stop-in), an error reduction could be attained regardless of kinematic variables. However, as the requirement of control increased (cross-over), a multi-variate adaptive strategy became necessary, thus leading to a speed-accuracy trade-off adjustment.

Decisional Correlates of Motivation

In line with the analyses of movement, our prediction was that social pressure would equally influence decisions between actions implying opposite biomechanical cost[33]. Since decisions were binary (right/left movement), our analysis was initiated by first fitting a binomial distribution to the proportion of times each participant chose the right T1 target over the total number of choices for each biomechanical arrangement (T1M/T1m) and experimental condition, and then fitted a GLM to the binomial p-parameters obtained at each experimental condition, was a function of Motivation (M), habituation factors (#Block, #Trial) and motor cost (BM×C) – see METHODS. We grouped together movement of similar cost by combining movements of both biomechanical arrangements: small cost were BM×C = 0; T1M×T1 & T1m×T2 movements, and large cost were BM×C=1; T1M Arrangement, T2 Choice & T1m Arrangement, T1 Choice. This grouping was possible because the arrangements were designed to reverse the costs of right and left movements when switching between arrangements (T1M to T1m[26,28].

FIG 3A shows the GLM group average regression coefficients and standard error, obtained from the binomial p-parameter for the *stop-in regime* reporting an effect of motor cost. This was signalled by the significant interaction between BM×C (p=0.00050, z=3.31). The negative coefficient indicates that higher

cost movements were generally preferred. Furthermore, as Motivation increased, choices were biased towards larger cost movements (M×BM×C, p=0.004, z=2.8; FIG 3B).

A significant interaction of biomechanics and choice (BM×C) was also reported from the same GLM analysis for the *cross-over regime* (p=0.039; z=-1.62; Suppl. FIG 1F). Unlike for stop-in, decisions in this case also adapted over time (#Block; p=0.004, z=2.8), were sensitive to fatigue (#Trial; p=0.0049, z=2.74) and exhibited a further influence of Motivation through two additional interactions: M×#Block (Supp. FIG 1G), M×#Trial (not shown). Remarkably, a visual comparison between the main effect and their interactions with motivation, shows that the influence of motivation by increasing social pressure for all three interactions reverses the influence of each factor. For example, if #Block decreased the likelihood of choosing T1, the modulation exerted by motivated state (M×#B) increases that likelihood. Finally, as social pressure increased from Solo to Easy and to Hard (M×BM×C), subjects' preferences transitioned from selecting the low to the high-cost movement, thus devaluating motor cost to gain a better precision (Suppl. FIG 1C).

Oculometric Correlates of Motivation

We monitored pupil diameter to assess whether our manipulation of motivation could be traced independently at a physiological level[34–38]. To test this, we pulled together the pupil diameter data across both regimes (stop-in & cross-over), z-scored each participant's recording, and averaged across two 1s intervals of interest, preceding two task events: the time the first origin cue was shown (tShowOrigin; tSO) and the GO signal (tGO) — see Methods, traces of two subjects in FIG 4A. We then quantified their dependency with respect to motivated state, #Block, #Trial, motor cost and their interactions with our GLM analysis (see METHODS). FIGs 4B-C report the resulting GLM group average and standard error regression coefficients for both intervals of interest. Two main group effects were reported: a significant pupil diameter increase with Motivation (F-test, F(11,1)=3.25, p=0.027 around tSO / F(11,1)=4.51, p=0.018 around tGO), and a reduction, possibly associated to fatigue, as the #Block increased (M×#B interaction, significant only around tGO, F(11,1)=2.95, p=0.032). Furthermore, we also assessed statistical significance by reiterating our GLM fit at intervals of 30ms, during both intervals of interest, and re-calculating a sliding t-test across group GLM regression coefficients. Remarkably, these yielded p-values < 0.05 over both intervals of interest (FIG 4D-E), demonstrating a consistent, baseline increase of pupil diameter caused by social pressure, which extended over most of the trial duration.

Electro-Encephalographic (EEG) Correlates of Motivation

Furthermore to assessing the influence of social pressure on motor decisions and pupilometry, our analyses of electro-encephalographic data aimed to identify the cortical network responsible for the changes of behaviour we reported. In other words, to asses the network responsible for Motivation as defined in this study, we adapted techniques previously used in fMRI studies[22] to our EEG recordings. We focused on the EEG activity recorded at an interval of 1.2s preceding movement onset for each trial, as to prevent movement artefacts and to focus on baseline effects during single block performance of brain

activity —keep in mind that Motivation and the control regime were the only constant experimental factors within block. The data was pre-processed by means of an ad-hoc pipeline for de-noising, participant-wise normalization across both sessions, and band-passing into the typical EEG a, b and g frequency bands — see METHODS. Our analyses combined two complementary levels of description: local electrode power activity and functional connectivity as a proxy for pairwise interactions in the electrode network and quantified by correlations (equivalent to spectral coherence within each band).

Our first set of analyses assessed whether and how three levels of Motivation (0-Solo, 1-Easy, 2-Hard) correspond to distinct brain activity states, as quantified by the electrode signals (FIG 5A-B). This we tested by a decoding scheme based on two usual machine-learning classifiers: multi-layer logistic regression (MLR) and k-Nearest Neighbour (1NN with k=1). For classification, we used the two sets of local/network features mentioned above (electrode power and pairwise electrode correlations) within each frequency band. Supplemental FIG 2A-B shows the classification accuracies obtained for a typical participant (Subject #2) and for the group, color-coded as a function of the frequency band. Remarkably, its accuracy increases from 65% in the a-band to a 95% in the g-band, when using the matrix of electrode power to classify. This held across participants, with remarkable consistency also for correlations, with a success rate from 50% in the a-band, to 85% in the g-band when using electrode pair-wise correlation as feature. On all accounts, the MLR outperformed the 1NN classifier by 5-10%, indicating that the activity states were better characterized by changes in specific features rather than by global changes of activity. A more detailed account of the state-by-state classification is also provided by the confusion matrices (Suppl. FIG 2A-B), which confirms the identifiability of three distinct motivated states for the higher frequency β- and g-bands, which is consistent across participants and local/network metrics.

We also identified the informative electrodes that support this robust classification, corresponding to electrodes whose power or electrode pairs whose correlation varied most significantly across the three motivated states. In this fashion, we gained access to the electrode network responsible for modulating motivation in our task. We relied on the recursive feature elimination (RFE) algorithm, which provides a ranking of how informative (or relevant) each feature is for the classification (see METHODS; Suppl. FIG 2C). Supplemental FIG 2D displays the most relevant electrodes, and the links between pairs thereof corresponding to correlations involved in modulating motivation. It points at changes in neuronal activity captured by frontal electrodes, but with connections to parietal electrodes.

Neuronal Correlates of Motivation

Beyond our analyses on electrode space, our second set of EEG analyses aimed at identifying the contribution of neuronal sources on the cortical surface, following the robust decoding and the distribution of changes across electrodes shown in Suppl. FIG 2. The classification procedure in the source space was analogous to the one just described, transposed to the source signals obtained from using independent component analysis on the pre-processed data[39,40]. We used a generic pattern to estimate the electrode anatomical locations provided for the 60-electrode Acticap configuration to

localize each source. Sources were obtained for each participant by pulling together both sessions and control regimes. Note that the source space had a reduced dimension of 40-50 after eliminating noisy and artefactual sources.

Reinforcing our results at the electrode space, the level of Motivation could also be decoded in source space for each participant and at the group level (FIG 5C-D). Again, accuracy increased with frequency, reaching 95% for the g-band, and ranked similarly regardless of metric (source power or correlation) per participant. Overall, the decoding performance and identifiability of the motivated states (see confusion matrices) were similar both in electrode and source spaces.

To identify the top relevant sources contributing to the classification, we performed RFE on the power sources. In brief, eight sources sufficed to attain a classification plateau of about 95% for all participants in the g-band (see for example Suppl. FIG 3). From those eight sources per participant, we identified the common sources that generalize across participants, pooling all sessions and control regimes across participants (two sessions per subject, two control regimes).

We used the Girvan-Newman community detection procedure based on spatial similarity on the cortical surface (see METHODS). This resulted in six common sources (communities of best sources), whose average spatial profiles are displayed in FIG 6A for each frequency band. In essence, these head maps correspond to the centroid of sources in each session belonging to the same community. We obtained sources in the prefrontal / premotor/ occipital cortices for a, b and g bands (FIG 6A). This broad distribution of sources related to motivation contrasts with the results of RFE applied to electrode power in Suppl. Figure 2D, which mainly identifies prefrontal electrodes. This also suggests a more complex network of processing involving not only prefrontal related areas, related to reward processing, but also premotor and visual cortical areas, consistent with the specifics of our experimental design.

Lastly, we verified whether the interactions across these common sources were modified by Motivation. To that end, we recalculated the classification performance using as features the correlations between the common sources per session alone. This is equivalent to selecting within the correlation matrices, the elements corresponding to the common sources obtained using community detection, akin to a massive subsampling. It is worth noting that a good performance was not guaranteed here because the best sources were identified on the grounds of their power alone, regardless of their cross-correlation (see the chance level performance of interactions within the alpha band in FIG 6B). Nevertheless, the accuracy turned out to be significantly above chance, for both b and g bands (FIG 6B). These results thus strongly suggest that these common sources also experienced network communication across distant brain areas induced by Motivation, highlighting the roles of b and g rhythms for this cognitive function.

Discussion

Here we studied how social pressure influences movements and decisions between them [41,42], and analyzed the network of cortical sources supporting this process. We introduce a novel experimental

protocol of decision-making between precision reaches using social pressure to manipulate intrinsic motivation in a controlled fashion, while bypassing the use of explicit external reward. Our participants performed either alone, or in the presence of virtual partners, capable of higher/lower accuracies than theirs. Participants were instructed to focus on their own performance and to disregard their partner's. Our results reported increased aiming accuracy (reduced error) whenever a partner was present, regardless of their skill. However, the participant's accuracy increased further when the partner was more skilled than her/him. At a physiological level, accuracy improvements were encompassed by an increase of pupil diameter. We quantified the influence of social pressure on movement under two regimes of control: stopping at the target (stop-in) or hitting and crossing over it (cross-over). Remarkably, our results show that, under these conditions, improved accuracy often favoured movement intensity, thus breaching the speed-accuracy trade-off typical of motor adaptation. We also used the participants' electro-encephalograms (EEGs) to characterize changes on the cortical network for each level of social pressure. In brief, we examined patterns of power activity and functional connectivity within the a, b and g bands for each participant, both at the electrode and source levels. Our results show that the MLR classifier could robustly categorize trials from each participant according to their Motivation state with a 95% accuracy in the g-band, both in the electrode and source spaces. We identified six common cortical sources that generalized across participants, mainly localised in the frontal, motor and visual cortices. Notably, the interaction across these common sources, achieved over 90% categorization accuracy in the g-band and 70% in the b-band, indicating motivation specific communication across these cortices. In summary, these behavioural, physiological and neuronal results point towards a transversal physiological effect, that is, an internal physiological drive that influences decisions and motor executions in the absence of explicit external reward.

From social pressure to intrinsic motivation and motor control

Social pressure is well-known to influence behaviour, by seeking outcomes that avoid being penalized [43] and achieve social success [24,44]. In our experimental design, each participant subjectively experiences success or failure at each trial across two full sessions, reporting two fundamentally different behavioural responses: when playing Solo, and when playing alongside a partner. Even under a condition of explicitly discouraged competition, and even if the partner is virtual, our experimental results (error reduction, peak velocity modulation, pupil diameter magnification, identifiability of separate brain states) are consistent with our participants experiencing some concern for performing worse than their partner, which yields better accuracies whenever a partner is present. However, the behavioural component of this adaptation can be along one of two strategies, always aimed at attaining that bit of accuracy that would equal or outperform the partner: a speed-accuracy trade-off and a cognitively controlled strategy.

In other words, if the goal of each movement is success, defined as attaining above average accuracy or a lesser error than the partner, the dynamics of a speed-accuracy trade-off adaptation dictate that going slower should increase your chances of reducing your aiming error [30]. However, our results show that this occurs only when playing Solo in the stop-in regime and essentially in all conditions for the cross-over regime. Our interpretation is that the participants improve accuracy by slowing down when there is no

external agent (Solo; Stop-in), and when the difficulty to improve precision is extreme due to signal dependent noise (Cross-Over). However, when there is social pressure and the difficulty to increase precision are moderate (Easy, Hard; Stop-in), the general trend is that of increasing precision and vigour simultaneously. Although further research will have to be conducted to ascertain this, this range of behavioural responses to social pressure, as a function of motor difficulty, suggests that adaptation within contexts of moderate difficulty — those in which motor noise is low, hinges on a cost-benefit rather than speed-accuracy trade-off [4], which may be cognitively controlled precisely because motor noise is within manageable boundaries. Under these conditions, the potential advantage is that of adapting by varying a single motor dimension, either amplitude or time, in a fashion independent of each other [45]. By contrast, as noise increases beyond some threshold, the only possible option to increase accuracy is to trade-off speed to increase accuracy.

One of the major rigors to claim that social pressure is a motivating factor is the requirement of attaining either a direct metric of its underlying physiological dynamics, or at least of a proxy thereof [2]. We argue that this proxy may be pupil diameter, which has been previously reported to co-vary, as a function of the experimental setup, with physical and mental effort, with arousal[35–38], and also with coping or with success[46]. The reason is that, in our experimental design, the motivating effect should transpire from the subliminal intention of 'making it' whenever accompanied by a partner, an effect that spills into the motivation related vigour effect and error reduction, and is consistent with the increase of pupil diameter we report. Our analyses of pupil diameter extend throughout the entire trial duration. Remarkably, our results show that pupil dilation positively co-varied with our manipulation of social pressure during entire trials, and for as long that specific block of trials (and exposure to the same type of partner) extended. In brief, this confirms the modulation of internal motivation regardless of movement metrics, and that an increase of social pressure is also consistent with a systemic rise of arousal. Beyond this motivational effect, we also observed an effect of fatigue on pupil diameter, which exerted an opposing depressing influence over time.

Motivated Motor control and Decision-Making

In our experiment, increased social pressure yielded a consistent improvement of precision. While this befits the participant's intent to outperform the partner, his/her adaptive strategy varied as a function of the context. As for the case of subliminal reward processing, we observed movement invigoration when the partner was less skilled than our participant during the stop-in regime alone. By contrast, if the partner was more skilled than the participant or during cross-over trials in all cases — arriving to target around peak velocity, subjects switched to the more cautionary strategy of weakening movement intensity and extending the movement, possibly to control the influence of signal-dependent noise on variability[29] and to ensure higher accuracies.

This introduces two main implications: first, although a low-skilled partner during stop-in trials exerts social pressure, it should not be that concerning, as the range of accuracies is such that participants may simultaneously increase velocity and accuracy. Importantly, these simultaneous changes are at odds with

the well-known speed-accuracy trade-off typical of the goal directed movements[30,47–50], suggesting that the dynamics of motivation may be, to some extent, cognitively reduced if properly motivated, and if the participant's skill allows[51]. Second, this effect is absent during the cross-over regime, possibly because of the intrinsic difficulty of controlling precision during ballistic, which overbears the capacity of cognitively controlling variability. The consequence is a more cautionary behaviour, consisting of longer time-to-peak-velocities (until target contact), and reduced velocity intensities when presented with a more skilled partner.

Moreover, consistently with the notion that decisions between actions abide by the same principles of motor control[33,52,53], our analyses of the pattern of decisions between movements yielded two main effects related to effort and motivation. First, decisions between movements were consistent with the participant's sensitivity to motor cost, typically preferring low cost movements in the Solo condition[26,27]. Second, motivation decreased the participant's sensitivity to motor cost and increased the likelihood of selecting high cost movements as the demand for end-point precision increased, consistently with the need of finding a path where to be more accurate [28]. In light of cost-benefit assessments[54], it could be interpreted that increasing accuracy is an abstract form of reward, that may be traded-off with the larger energy expenditure required of more stable movement paths. Future research needs to be conducted to clarify this point.

The cortical network of intrinsic motivation

In addition to the modulation of pupil diameter, we also analysed the participant's EEGs to characterize the activity and the communication across the related brain sub-networks typical of each level of social pressure. Our EEG analyses focused on a time interval preceding the presentation of both movement options and the decision-making phase. Consistent with this, we did not find the usual a-band occipital modulation typical of sensory tasks increasing attentional load[55], but rather an increase of coordination, best reported in the g-band. We pursued two complementary levels of analysis: local power and network-wise functional connectivity, in both electrode and source spaces. In line with the EEG literature, we band-pass filtered the signals into three typical frequency bands: a [8-12 Hz], b [15-32 Hz] and g [40-80 Hz]. These rhythms have been traditionally associated with specific functions, in particular with visual and motor processing: 1) the a-band with visual attention and surround suppression for stimuli within the visual field[56]; 2) the bottom-up processing associated with g-band and top-down processing associated with b-band in the visual hierarchy[57]; 3) the b-band power reduction over the motor cortex to signal motor readiness and forward control[58]; 4) the g-band with feedback control[59,60]; 4) the a-b coupling with hand grip tasks[61].

In our task, because of the necessary visual movement planning and execution, and coordination at different levels, we expected motivation to reflect differently at each frequency band. The strongest modulation (matching the highest classification accuracy), both locally with source/electrode power or via network interactions — quantified by functional connectivity, was mildly observed in the b-band, and

more strongly in the g-band (FIGS 5-6 & Suppl. FIG 2). This contrasts with the a-band modulation associated with bottom-up attentional processes in the primary visual cortex[55], as well as with the bottom-up and top-down communication related to g/b bands in the visual cortex[57].

As to analyze the structure of the network differences from baseline onto motivated states, we performed an analysis of source contribution to each accuracy calculation (RFE analysis across subjects), revealing that several distant cortical regions (FIG 6), contributed to this network, exhibiting similar spatial patterns across frequency bands with various levels of intensity. Consistently with previous studies, we found pre-frontal cortical sources responsible for the modulation of motivation, suggesting the engagement of assessment processes and some of the neural circuitry responsible for reward processing[13,62,63]. Likewise, we identified sources in motor cortical sources (located in central regions), consistent with the nature of our task. Finally, the involvement of sources within the parietal and visual cortices, points at multi-modal integration, an introspective state [64], and the baseline modulation on visual attention. Together, our results hint at a distributed cortical network effect engaged by intrinsic motivation, which exceeds pre-frontal localization often associated with the encoding of reward and also involves cortical areas naturally associated in our visuo-motor task.

Limitations and Future Work

Our brain network interpretation is necessarily constrained by the limitations of EEG recordings, as more detailed head models with higher EEG density would be required for a finer spatial source localization[65]. Furthermore, the fact that we found physiological correlates of motivation on pupil dilation and that the g-band is the one the most affected by motivation (and not in the a-band), would suggest that a process other than attentional load is responsible. Further experimental work should be conducted to confirm this. Finally, our work is constrained by a model-free analysis of connectivity, which could be extended by a more realistic bio-physical modelling of brain dynamics and communication, as recently achieved for fMRI signal analyses[66]. This would also allow to interpret cross-band communication in a unified framework. Ultimately, our goal was to entirely disentangle motivation specific from reward specific cortical communication. In addition to the proper analytical tools, this would require a novel experimental context where neural activity may be compared between two experimental conditions, one where explicit reward and motivation were present, and another where motivation alone drives behaviour. We leave this for a future study.

Conclusion

We showed that motivation may be manipulated by social pressure in the absence of an explicit external reward. Furthermore, this modulation is reflected in neuro-physiological signals that leads the way to the design of sophisticated future experimental work aimed at the careful disentanglement of the neural dynamics of motivation, reward and attentional processes in a quantitative manner.

Materials And Methods

Participants

Eleven right-handed individuals (4M and 7F; Mage = 55yrs, SD = 5.8), participated in this research study. All participants had normal or corrected-to-normal vision and did not suffer from any known neurological disorders. Informed consent was obtained following the guidelines established by the local ethics committee and all participants received monetary compensation (10€/h) for their participation, regardless of completion. The ethics protocol for this study was approved by the Clinical Research Ethics Committee (CEIC-Parc Salut MAR) of the Pompeu Fabra University-Hospital del Mar, Ref. #2015/6085/I. All methods were carried out in accordance with relevant guidelines and regulations.

Experimental Setup and Apparatus

Participants were situated in an electrically shielded room of the neuroscience laboratory (Center for Brain & Cognition, Pompeu Fabra University), where the task was performed and ocular and electro-encephalography (EEG) was registered. The experimental began after electrode preparation. The participants were seated in a comfortable chair, facing the experimental table, with their chest approximately 10cm from the table edge and both lower arms resting on its surface. The table defined the plane where reaching movements were to be performed. On the same table, approximately 60cm away from the participant's sitting position, we placed a vertically-oriented, 24" Acer G245HQ computer screen (1920x1080). This monitor was connected to an Intel i5 (3.20GHz, 64-bit OS, 4 GB RAM) computer that ran custom-made scripts which controlled task flow, programmed using OpenFrameworks v.0.9.8 software. The screen was used to show the geometrical arrangements and related stimuli on each trial. A small cross (1x1cm), whose position was synchronized with the planar coordinates of the end-point as it slid along the horizontal plane (table), was used to show the participant's corresponding movement in the vertical plane on the screen.

As part of the experiment, subjects had to respond by performing overt movements with their arm along the table plane. Their movements were recorded with an Optitrak motion tracking system (Optitrak, Inc; Corvallis, OR, USA) sampling at 100Hz, which tracked the position of a spherical marker placed on the nail of the right-hand index finger, as it slid on the table plane. We performed a spatial calibration to synchronize the axes and units for movements on the table Tracked by the Optitrak system with those of our custom-made software. The marker position was synchronized by the custom-made software controlling the task flow. Subjects were instructed to maintain end-point contact with the table surface at all times, and to apply minimal pressure. They were permitted to minimally lift their elbow off the table to diminish this effect. Given that the monitor was placed vertically and movements were performed horizontally (along the table surface), the movement component along the sagittal plane was rotated 90 degrees, to the frontal plane, to show displacements of the end-point on the screen. The transverse movement component was shown unaltered. Data analyses were performed with custom-built MATLAB scripts (The Mathworks, Natick, MA), licensed to the Pompeu Fabra University.

The subject was required to maintain posture at a fixed distance from the table and to place his/her chin on the chinrest. Eye gaze and pupil diameter from both eyes were tracked and recorded with an EyeTribe

oculometer (Oculus, Menlo Park, CA, USA), sampling at 60Hz. We used a chinrest to stabilize posture and fix the head position at a predetermined distance from the screen and from the oculometer. The signals delivered by the oculometer were recorded by the OpenFrameworks custom-made code, along with the movement trajectories and other behavioural data. Behavioural, ocular and electro-encephalographic data from each session were transferred to a MySQL community server database (Oracle, Redwood Shores, CA, USA) for further analysis using custom-designed MATLAB scripts (Mathworks, Natick, MA, USA).

The electro-encephalogram (EEG) data were recorded from 60 points on the scalp by using a high-density 64 active electrode actiCap Brain Product, amplified by a Brain Product amplifier (Brain Products BmbH, Gilching Germany). An appropriate head-sized cap was selected, and conductive gel was carefully applied to obtain electrode impedances <5 kW. Impedances were checked every other block to maintain this condition. Eye movements were measured with electrodes attached to the infraorbital ridge and on the outer canthus of the right and left eyes. Electrode impedances were kept <5 kW, and we used a central electrode voltage reference. The electrophysiological signals were filtered with a bandpass of 0.1–100 Hz and digitized at a rate of 500 Hz. EEG recordings were performed by an independent computer (Intel i5; 3.20GHz, 64-bit OS, 8 GB RAM) running Brain Vision on Windows XP. External pulses, generated by the custom made OpenFrameworks code, were used to synchronize the recordings from both computers on a single trial level.

Experimental Task

The participants performed a decision-making task between two reaching movements, seeking precision at target arrival. Maximum precision is attained when arriving at the centre of the wide side of one of both rectangle targets. To provide a positive metric of precision other than the error, we defined a metric of reward that decreased linearly with the error (FIG 1B). During the experiment, we varied three factors: motivation, biomechanics, and the requirement of stopping at the target (FIG 1B-C). The parametrization of the two first factors is explained in the sections *Manipulation of Internal Motivation* and in the section *Manipulation of Biomechanical Factors*. The constraint of stopping was implemented by defining two motor control regimes: stop-in, instructing the subject to stop at the target, and cross-over, instructing the subject to punch through it and to stop whenever afterwards.

There were twelve experimental conditions, as a function of three main experimental factors: two geometrical arrangements (T1-Major, T1-Minor; FIG 1C), three motivated states (0-Solo, 1-Easy, 2-Hard; FIG 1D), and two endpoint control conditions (Cross-Over/Stop-In; FIG 1C). According to these, we designed 10 blocks of 108 trials each; blocks 2, 4, 6, 8 and 10 were Cross-Over, and blocks 1, 3, 5, 7 and 9 were Stop-In. Furthermore, blocks 1 and 2 were of type Solo, 3, 4, 7 and 8 were of type Easy, and 5, 6, 9 and 10 were of type Hard. At each block, the number of T1-Major vs T1-Minor trials within block was the same (50 each); we incorporated 8 single-target trials per block to encourage the participants to equally experience all movements. For each participant, we interspersed the 8 non-Solo blocks into two sessions, maintaining 2 Cross-Over and 2 Stop-In per session, and including blocks 1 and 2 in both sessions to

have a baseline motivated state for each session. Both sessions were performed in consecutive days for each subject.

Real-time, visual feedback of hand position was provided throughout the trial by a 1cm cross displayed on the screen. Its position was synchronized with that of the participant's right index finger sliding on the table. The time-course of a typical trial is shown in FIG 1A. Within each trial, both potential trajectories were defined by the origin cue (pale blue dot; radius 1cm), a via-point (pale-blue dot; radius 1cm), and a target (dark-blue square; side 10cm, depth 1cm — FIG 1A). Each trial began when the origin cue was shown on the screen (FIG 1A) — the Time Stimulus Onset (tSO). Then, the subject slid his/her right-hand index finger into the origin cue. After holding position there for a 500ms — Origin Hold Time (OHT), the geometrical arrangement (via-points and targets) defining one or two potential trajectories was shown. After a 1000ms Observation Time (OT), a GO-signal was given (origin cue disappeared). The instruction was to react as fast as possible, to freely choose an action, and to be as precise as possible by sliding over the via-point and towards the target while maintaining the index fingertip in contact with the table surface at all times. 100ms after entering the target, the participant was informed of the precision attained by a green bar. The bar length ranges between 0 and 900 screen pixels, and its size is scaled between zero if the error is equal or larger than the side of the rectangle, and 900 if the error is zero. Any intermediate value scales the bar down linearly. If accompanied by a simulated partner, a red bar is also shown, indicating the performance of your partner in the same trial. Finally, the ranking of the participant vs the partner is shown each nine trials during 500ms, after the precision bars. Subjects gave informed consent to be photographed for this purpose only. Their photograph was shown alongside that their partner on a vertical axis of precision. No photographs were kept post-experiment.

Manipulation of Motivation

Our manipulation of motivation was performed by means of a social hierarchy defined within the experiment, as a function of the participant's aiming skill with regard to that of simulated partners. We informed the participant that he/she would perform within a community of participants, and that at each block, he/she would have a different partner from this community to perform alongside. To reinforce the belief of a real partner, at the end of each trial we showed a horizontal green bar displaying the accuracy attained, normalized between 0 and 100%, alongside a red bar displaying the accuracy of their partner. Each nine trials we also showed a photograph of the participant and of their partner, ranked in a top-down fashion, at a height proportionally to their average end-point accuracy. Since the goal of the simulated partner was to introduce a subliminal bias that would modulate the participant's intrinsic motivation, we informed each participant this was not a task of competition, that the partner was for companionship purposes, that the ranking was only to keep track of your performance, and that the participant should focus only on performing the task and not on outperforming their partner. To parameterize the bias introduced by the presence of the virtual partner, we classified partners of two types, as a function of their worse of better aiming accuracy with respect to the participant (FIG 1D). We also matched the participant and the partner gender to avoid cross-gender effects.

Despite the instruction, we predicted that participants will be concerned by their accuracy with respect to their partner, and that they will adjust their process of selection of motor parameters and/or their policy to select between movements. However, this could be attained by simply trading-off speed by accuracy, or by varying one of these two metrics in a cost-benefit scenario. The influence of the motor cost on the choices was varied in a random fashion on a single trial basis, as we used one of two geometrical arrangements (FIG 1A). By contrast, the social position and the control regime were maintained constant during each block and varied across blocks (10 block types), presented in a counter-balanced fashion. Each participant performed two sessions of six blocks of one-hundred and eight trials each. Each session included six blocks, three per control regime; one block solo, and two blocks alongside both kinds of partners. Subjects were given real-time visual feedback on their trajectories.

Criteria for a trial to be considered faulty were: if the position of the endpoint left the origin before the GO signal, if the reaction time (RT) was shorter than 200ms or longer than 1000ms, or if the stylus reached the target before first crossing over the via-point. Each individual trial was shown to the subject a single time. In other words, even if a trial were faulty, the sequence of trials resumed to present the subject with the next trial. Error trials in which the subject crossed the via-point (where a significant part of the trajectory was however performed) but failed to enter the target or failed to stop at the target during the THT time were disregarded from further analysis. Visual feedback was provided during the movement by showing the end-point position as a small cross in real time. Furthermore, the colour of the via-point and target cues changed to green as tip of the small cross slid over them.

Behavioural Analyses

Each end-point trajectory was first low-pass filtered at 10Hz. We then identified and quantified the following kinematic markers for each individual trajectory: Peak Velocity (PV), Time-to-Peak Velocity (TTPV), Movement Time (MT), Peak Acceleration (PA), Time-to-Peak Acceleration (TTPA) both for the stop-in and cross-over regimes, and Peak Deceleration (PD), and Time-To-Peak Deceleration (TTPD) for the stop-in only regime — FIG 1D, and were converted into z-score for statistical analyses (for each subject individually), and performed a Greenhouse-Geisser correction for non-sphericity. As a function of our hypotheses of control for each control regime, we then fitted a generalized linear model (GLM) on a single subject basis for several of these metrics. For the stop-in regime this comprised the late kinematic markers, involved in the control of stopping: PV, TTPV, PD, TTPD; and for the cross-over regime the early kinematic markers: PA, TTPA, PV, TTPV, as target arrival occurred around peak velocity. The GLM included the following variables: motivated state [0, 1, 2], #Block [1, 2, 3], #Trial (within Block — [1-108]), Biomechanical Cost (Biomechanical Arrangement — 1|2 (T1-Major|T1-minor) x Choice — T1|T2) and interactions between them, as defined by equation 1. Note that the BM×C, was define to group low-cost and high-cost movements within arrangement (0-Low Cost/1-High Cost), and to ultimately quantify the influence of motor cost on our metrics.

$$\phi^k = \beta_0^k + \beta_1^k M + \beta_2^k \#B + \beta_3^k \#T + \beta_4^k BM \times C + \beta_5^k M \times \#B + \beta_6^k M \times \#T + \beta_7^k M \times BM \times C \quad (1)$$

For each explaining factor, we identified the regression coefficients b that significantly differ from 0 (as reported in FIG 2A-B and Suppl. FIG 1A-B) by running a Mann-Whitney-Wilcoxon (rank sum) test between the distribution of b pooled over all subjects and a null distribution. The null distribution was built by shuffling the regressed values f^k across trials within each subject, then pooling the surrogate coefficients of each type across all subjects together. Note that the z-scoring applied to the kinematic markers allows for the alignment of the distributions across the different subjects. In this way we identified the explaining factors (and interactions thereof) that affected the kinematic markers (as quantified by the regression) to compare the effect of motivation with other expected factors like habituation or fatigue (#Block and #Trial) and biomechanical constraints (BM). We set the threshold for group significance at $p<0.05$ with the Mann-Whitney-Wilcoxon (rank sum) test to compare the distribution of regression coefficients estimated from the data and the corresponding null/surrogate distribution.

Oculometry Analyses

Like the movement markers, we also z-scored our recordings of pupil diameter and regressed this metric with the same GLM used to test behaviour (Eq. 1). We calculated the GLM b-coefficients per subject at each 30ms during two intervals, centred around the tSO and at the tGO events (FIG 4C-D). We determined group significance for each variable by running a t-test across the b-regression coefficients obtained across subjects (FIG 4A). Again, the threshold for statistical significance was set at $p<0.05$ (without correction for multiple comparison, considering each time step to be independent).

EEG Pre-processing

We selected an interval of interest of 1200ms for further analyses, starting 800ms before the first stimulus onset — the initial cue, and ending 400ms later (see Fig. 1A). We selected this interval for two main reasons, first to seek for baseline changes in the network of motivation, which should not depend on each specific trial but on the block of trials — the partner was the same during each block of trials, and so should the modulation of motivation. Second, this interval preceded any movement, and was therefore likely to contain fewer artifacts than those following the stimulus presentation.

We applied a 4th order notch filter around 50, 100 and 150Hz to prevent electrical interference from the power supply line and a 4th order bandpass Butterworth filter between 0.1 and 100Hz. Electrodes with EEG level exceeding either 200V or voltage step/sampling 50V within intervals of 200ms were removed off-line. Baseline was corrected, and the datasets were z-scored by using the recordings during trial block types 1 and 2 of each session — when the participant was playing solo, as a reference for baseline motivation.Eye related artefacts were removed by means of independent component analyses (ICA), implemented in with custom-made Field-trip open-source toolbox (www.fieldtrip.com) and EEGLAB scripts

(http://sccn.ucsd.edu/eeglab, UC San Diego, CA, USA). The procedure to identify eye-movement related sources was semi-automatized, first correlating each source obtained with the signal from the electrodes recording eye movements to obtain a first metric of relatedness. Second, we visually inspected all sources to corroborate that their shape and spatial location matched those of ocular artefacts. Eye-related sources were removed and the cleaned signal obtained by inverting the ICA process.

Extraction of Neuronal Signals (Source Space)

The pre-processed EEG datasets were transformed from electrode into source space via custom-made MATLAB scripts based on the default EEGLAB ICA algorithm in combination with the electrode spatial location map. The Brain Products Unicap 64 configuration was used to establish a spatial reference between the electrode placement and the sources location, and we assumed a spherical head model.

EEG Analyses in both Electrode and Source spaces

We used cross-validated classification to assess how much information about motivation is present in the EEG signals. Following the task description, we defined three motivated states as a function of the types of partner the participant performed the task with: Solo (M=0), when alone; Easy (M=1), when performing with a partner of lesser skill than the participant; Hard (M=2), when performing with a partner more skilled than the participant (see behavioural task, FIG 1D). The signals, in electrode or source space, were first filtered in three frequency bands: a [8-12Hz], b [15-32Hz], g [40-80Hz]. We cut out our interval of interest from the EEG temporal series, from 800ms before the first stimulus onset, until 400ms after (yielding 1200 time points at a resolution of 500Hz), i.e. at the tShowTarget event (FIG 1A). We applied our classification pipeline to each recording session, type of signal and frequency band. This pipeline relied on two specific metrics to capture complementary aspects of the brain network of motivation: signal power and pairwise correlation between signals (as a proxy for interactions, which is equivalent to spectral coherence averaged over each frequency band used here). The classifiers were trained to predict the Motivation Category with a train set (entailing 80% of the trials) and to yield a performance with a separate test set (the remaining 20% trials). This procedure is repeated 100 times with distinct train/test sets obtained by randomly splitting the 1296 total trials (432 trials / Motivation Category), to comply with the nowadays standards for proper assessment of the generalization capability of classification [67].

The code for classification consists of custom-written (open-source) python scripts with the numpy, scipy and scikit-learn libraries [68]. We used two classifiers: multinomial linear regression (MLR) and 1-nearest-neighbor (kNN with k=1), in a similar manner to a previous study for fMRI data [69]. It followed nowadays standards for proper assessment of the generalization capability of classification procedure [68]. Each classifier performed a distinct mapping from features to motivated category. The MLR calculates a linear weighted sum of the entries, which is then rectified by a sigmoid function, to calculate an output ranging from 0 to 1 for each category. This indicates the confidence of the classification in the predicted category; the decision is made by selecting the category whose confidence is the highest. The training procedure tunes the weights (or regressors) for all features to reduce the prediction error on the train set. By contrast,

the 1NN calculates a similarity measure between the sample in the test set and all samples in the train set (using the Pearson correlation between the two vectors of features) and attribute to the test sample the category of the most similar train sample. Here the 1NN does not involve training.

To identify the most informative (or “best”) features for the classification, we used Recursive Feature Elimination (RFE) on the MLR; because the MLR gives a better performance than the 1NN. In brief, its strong weights (in absolute value) correspond to important features (also because the MLR gives a better performance than the 1NN). RFE provides a ranking of features by their importance by iteratively excluding features that weakly contribute to the classification and optimising the remaining weights. Ultimately, the most relevant features survive the pruning process. In practice, we performed RFE for each train set (in each train/test split) and then evaluate the stability of the feature rankings by calculating the Pearson correlation coefficient between all pairs of rankings. In the case of stable ranking (typically Pearson correlation coefficients above 0.6), the mean ranking over all train/test splits is calculated. This gave two sets of best features: electrodes for power and interactions between pairs of electrodes for correlation (FIGS 5-6 and Suppl. FIG 2).

Common sources

The common sources were calculated for each participant using the power of source signals. Following the heuristic that the eight best sources obtained using RFE were sufficient to attain an over 90% accurate classification (FIG 6B), we pooled the eight best sources over all sessions of all subjects, over both motion types. We used the Girvan-Newman algorithm [70] to calculate communities of similar sources (spatially distributed over the cortical surface) among the original 25-45 sources obtained per subject. Similarity between pairs of sources was established using the Pearson correlation over the vectorized head maps, after retaining only the more localized part (similar to a spatial rectification to favour narrow bumps). Once the sources were grouped in communities, the *centroid* head map of each community was calculated by averaging all corresponding sources. We performed this operation independently for each frequency band, yielding six communities of common sources for the a, six for the b-band and five for the g-band.

Declarations

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Figures

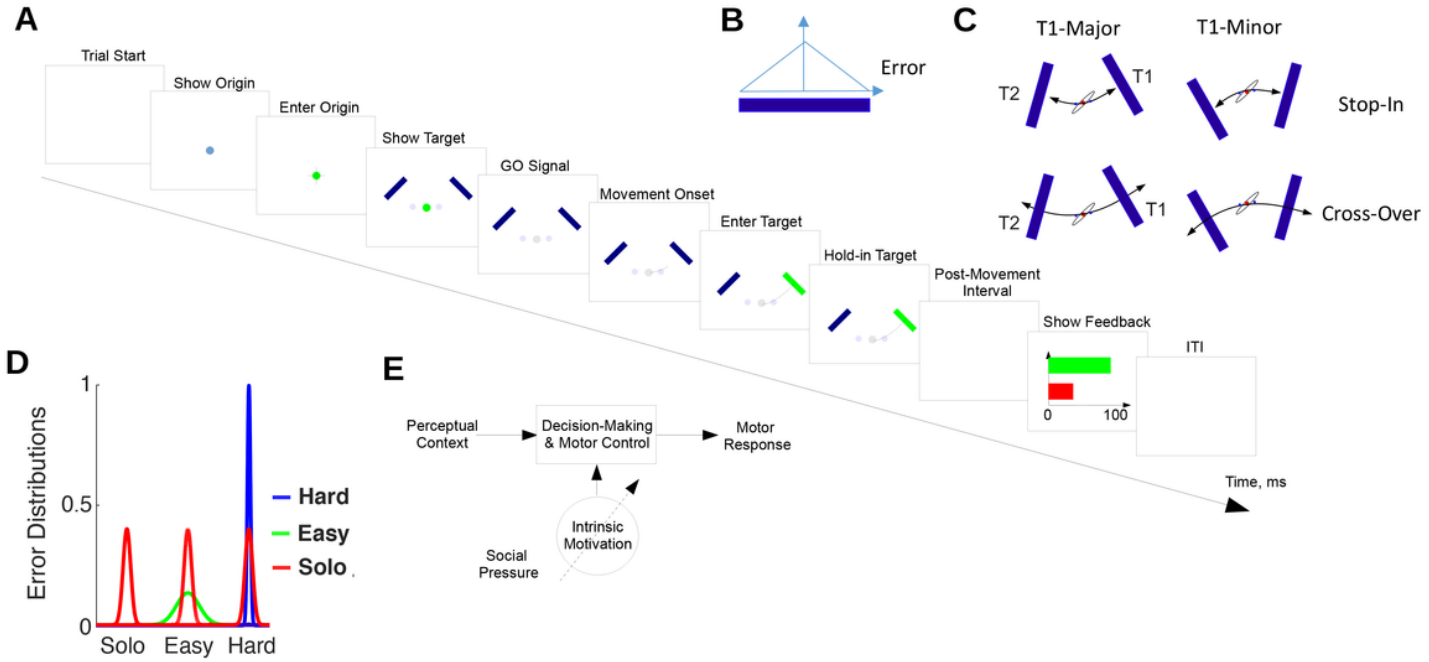

Figure 1

A. Timeline of a typical trial. It starts with the presentation of a white screen during 500ms. After this time, a 1cm diameter pale blue origin circle cue is presented on the centre bottom part of the screen. 1ms after the subject's fingertip crossed its border, the origin cue changed to green, and two rectangular blue targets (10 x 1cm) and two pale blue via-points (1cm diameter) we presented to the right/left of the origin cue. Path distances from the origin to either target extended 15cm. The GO signal was reported by removing the origin cue, at which time the participant could report his/her choice by moving the cursor from the origin to the selected target. The rectangle target turned from blue to green when the cursor entered the target. After 500ms, everything disappeared. 500ms after this, a green horizontal bar was shown during 500ms to report the precision of the movement just performed. If the participant was accompanied, a red bar was also shown below, reporting the simulated partner's precision. An empty screen ITI followed, the duration of which extended the trial to its 7000ms fixed duration. B. Triangular distribution of reward as a function of the error upon arrival to the target. The reward is 1 at the centre and decays linearly towards the edges of the rectangle. It is zero off the rectangle. C. Geometrical arrangements (T1-Major/T1-Minor) and control configurations (STOP-IN/CROSS-OVER). In the T1-Major arrangement, the reaching movement towards T1/T2 implies a small/large motor cost. D. Depiction of the distributions of error associated to our artificial partners (Green: Easy –- broad variance; Blue: Hard –- narrow variance), and of a participant (Red). E. Schematic of the interaction between social categorization, intrinsic motivation and the process of decision-making between actions and of motor control.

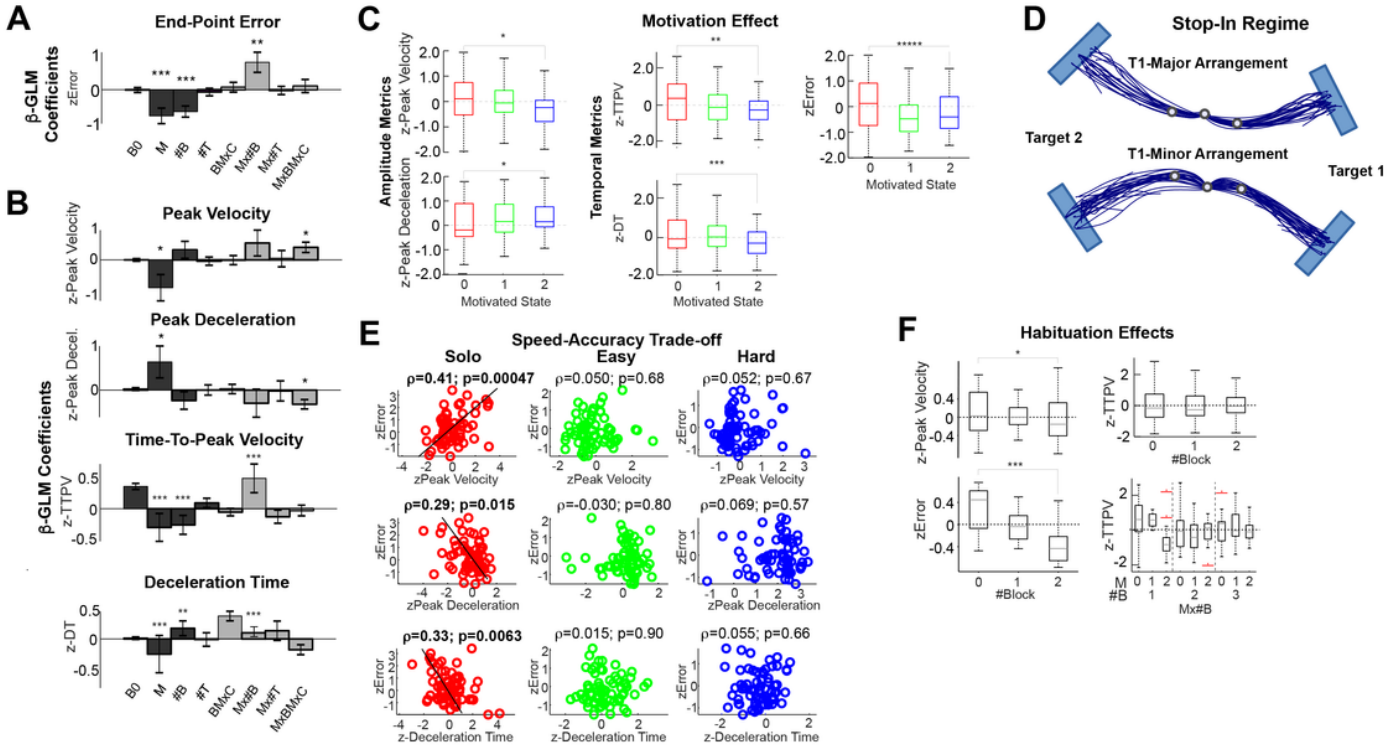

Figure 2

Behavioural Effect of Motivation for the Stop-in Target Condition. A. Group average (and std. error) GLM β-regression coefficients for the end-point error upon target arrival (see METHODS). List of dependent variables: B0 (intercept), M (motivated state 0-2), #B (#Block), #T (#Trial), BMxC (Biomechanics; T1-Major/Minor x Choice; 1-2), Mx#B (Motivated State x #Block), Mx#T (motivated state x #Trial), MxBMxC (Motivated State x Biomechanics x Choice). B. Same as A, but for peak velocity (PV), peak deceleration (PD), Time-to-Peak Velocity (TTPV) and Deceleration Time (DT) – see METHODS. C. Behavioural effect of the motivated state on the z-scored PV, PD, TTPV, DT and End-point Error. Their group average distributions have been boxplot for each motivated state (Red: Solo; Green: Easy; Blue: Hard). D. Behavioural effects of habituation, Both the error and the peak velocity diminish as a function of the #Block. There is also a mixed effect between Motivation and #Trial within block, which increases the error and peak velocity. E. Interaction effects between Motivation and Habituation Metrics. D. Average end-point trajectories for each Biomechanical Configuration (T1-Major/T1-Minor) and target choice (Red: Left, Blue: Right). E. Speed-accuracy group analyses. Scatter plots for for the end-point vs peak velocity (top row), peak deceleration (middle row), and TTPD (bottom row), for each motivated state (Solo: Red; Easy: Green; Hard: Blue). Top correlation and significance p-values have been displayed above each inset. Notice that the correlations typical of a speed-accuracy trade-off only hold for the Solo case, meaning that error reductions during the Easy & Hard cases do not co-vary with typical kinematic markers such as PV, PD & DT. F. Behavioural effects of habituation (as a function of the #Block per session) and interaction between Motivated State and #Block on the PV, TTPV and Error. Notice that the PV and Error diminish with #Block, consistently with a learning or habituation effect.

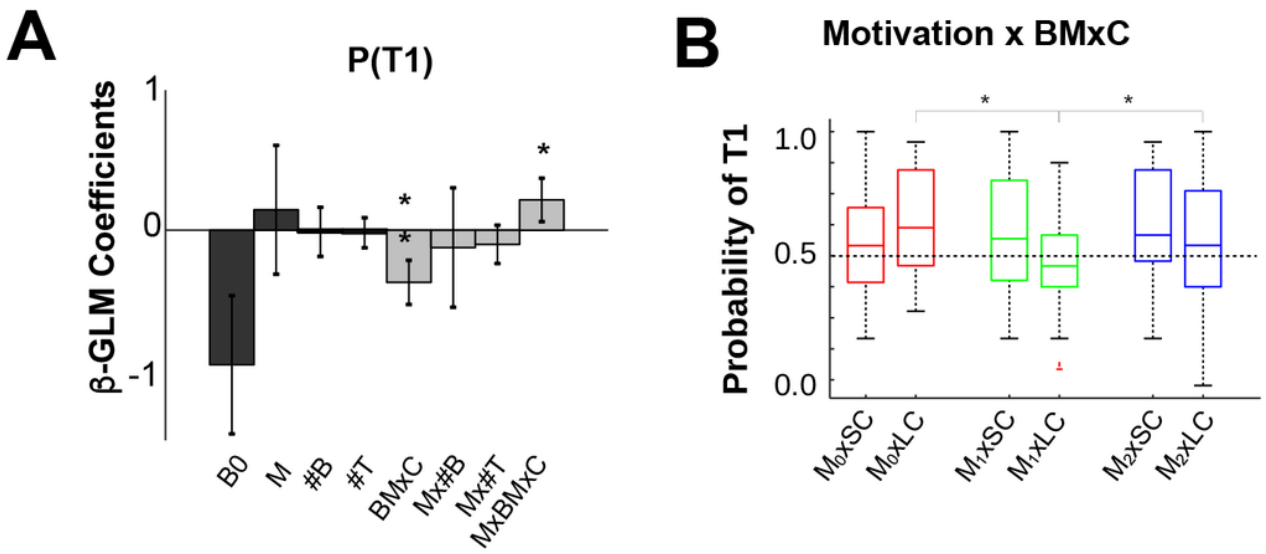

Figure 3

Behavioural Effects of Motivation on the Choices (Probability of Choosing the Right T1 target) in the Stop-in Configuration. A. Group average and standard error GLM ®-regression coefficients for the p-binomial parameter –- P(T1). List of dependent variables: B0 (intercept), M (Motivated State 0-2), #B (#Block), #T (#Trial), BMxC (Biomechanics –- T1-Major/Minor x Choice; 1-2), Mx#B (Motivated Stat x #Block), Mx#T (Motivated State x #Trial), MxBMxC (Motivated State x Biomechanics x Choice). B. Interaction between Motivation and Motor Cost (BMxC) for P(T1).

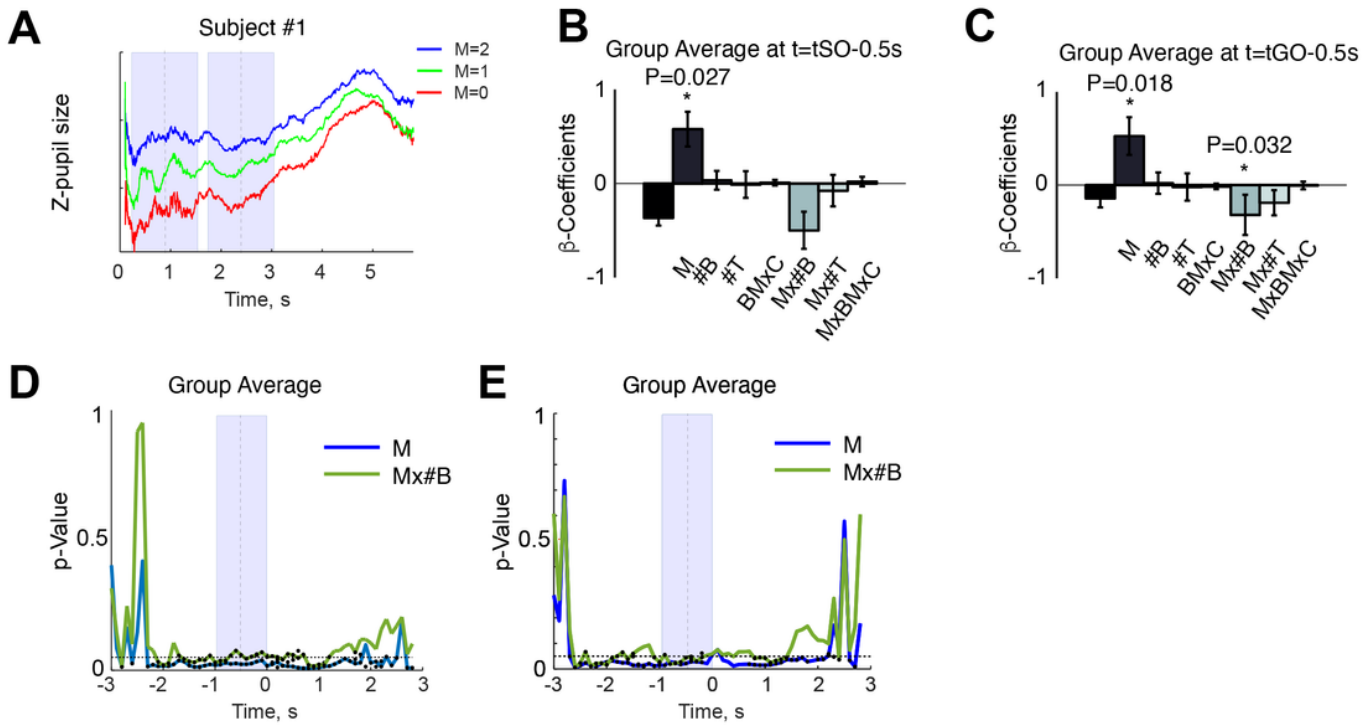

Figure 4

Analysis of pupil-size during performance of the motivated state Task. A. Sample average pupil diameter traces for S1 during the duration of the trial, colour coded as a function of motivated state (M0-Red; M1-Green; M2-Blue). B. Group GLM β-regression coefficients (mean and std. error) of the z-scored pupil diameter, aligned on the time of stimulus onset (tSO), and averaged over the 500ms preceding this event. The factors considered are: the motivated state condition (M), #Block (#B), #Trial (within Block) –- #T, BM x C (Biomechanical Configuration x Choice –– as a proxy of effort), and the interactions between the motivated state and the Habituation metrics (#Block, #Trial) and with BMxC. C. Same as B, but aligning our recording of the pupil diameter on the GO signal and averaging over the 500ms preceding this event. Significant effects are reported as * p<0.05.D. P-value resulting from the t-test over the M and Mx#B β-regression coefficients for case B, during a 3s interval preceding/following the tSO. The boundary of significance P<0.05 has been drawn as a horizontal black dashed line. The pale blue area indicates the interval considered to calculate the regression coefficients show in E. P-value resulting from the t-test over the M and Mx#B ®-regression coefficients for case C, during a 3s interval preceding/following the tGO. The boundary of significance p<0.05 has been drawn as a blacked dashed line. The pale blue area indicates the interval considered to calculate the regression coefficients show in B.

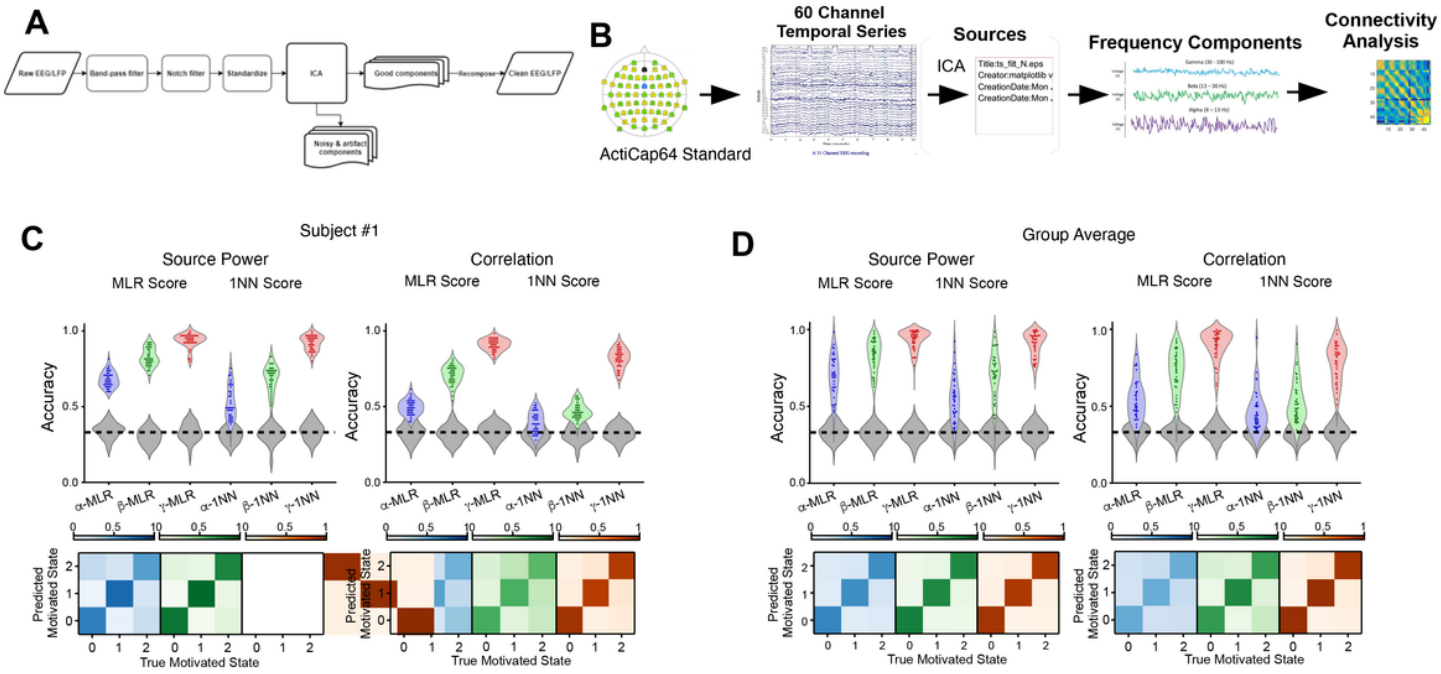

Figure 5

All Sources Space EEG Analyses. A. EEG pre-processing pipeline. It starts with a band-pass filter between 0.1 and 100Hz, a notch filter to remove power supply noise at 50Hz and its harmonics (100, 150Hz). This is followed by a de-composition into independent components to identify and remove eye and muscular components, to ultimately re-compose the original signal with the original components. B. Pre-processed EEG analysis pipeline. It first includes an ICA decomposition for the case the analyses were performed into source space (FIG 6). This is followed by a decomposition into the α, β and γ-bands, with are used for further analysis. C. Motivated State classification accuracy and confusion matrices, as by the MLR and 1-NN classifiers, as a function of #Motivated state, for each frequency band and metric (electrode power, correlation), performed on the EEGs of Subject #1. D. Same as C, averaged across all subjects.

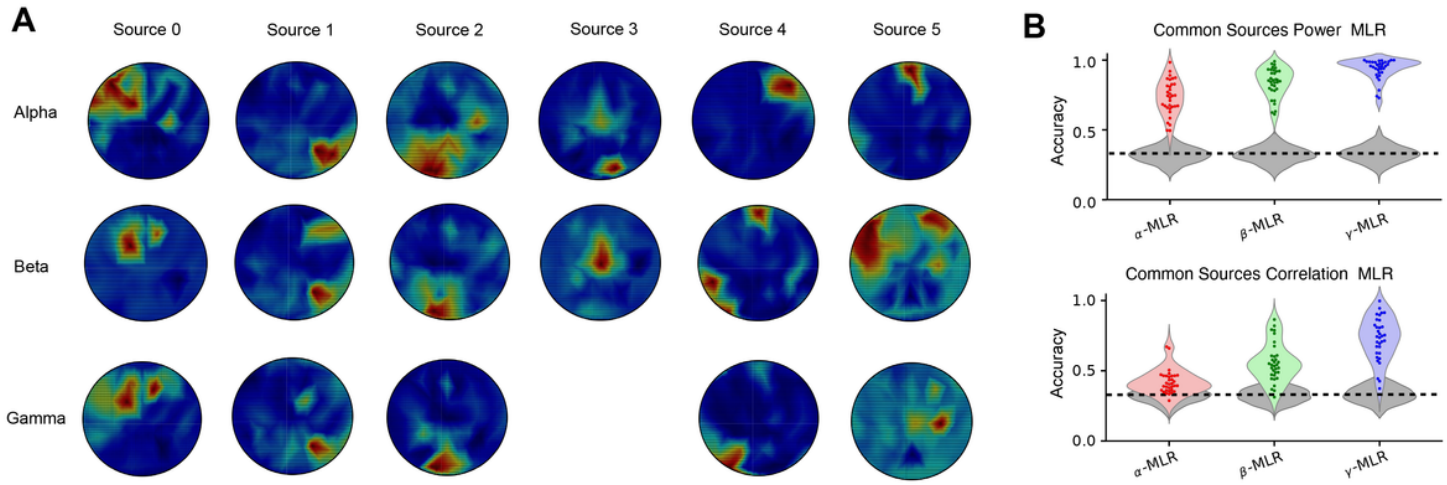

Figure 6

Selected Common Sources Space EEG Analyses. A. Common sources across subjects related to the task for the α, β and γ-bands . B. MLR Classification accuracy for each frequency and metric (source power, source covariance, source correlation).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- CosDecoGilsonFigs2020v28Suppl.pdf